

Calc 1 Lecture 6: Limits at infinity (horizontal asymptotes)

Observation: Some functions seem to approach a limiting value as x gets very large.

Example: $\frac{1}{x} \rightarrow 0$ as $x \rightarrow \infty \Rightarrow \lim_{x \rightarrow \infty} \frac{1}{x} = 0$

Example: Consider $f(x) = \frac{x^2 - 1}{x^2 + 1}$

What do you think happens to $f(x)$ as x gets very large?

intuition: Focus on dominant terms

$$\begin{aligned} x^2 - 1 &\approx x^2 \\ x^2 + 1 &\approx x^2 \end{aligned} \Rightarrow \frac{x^2 - 1}{x^2 + 1} \approx \frac{x^2}{x^2} = 1$$

guess: $\lim_{x \rightarrow \infty} \frac{x^2 - 1}{x^2 + 1} = 1$

More Precise explanation

$$\frac{x^2 - 1}{x^2} \rightarrow 1 \quad \left(\text{explanation: } \frac{x^2 - 1}{x^2} = \frac{x^2}{x^2} - \frac{1}{x^2} = 1 - \frac{1}{x^2} \right)$$

as $x \rightarrow \infty$ $\frac{x^2 + 1}{x^2} \rightarrow 1$ (similar explanation)

$$\frac{x^2 - 1}{x^2 + 1} = \frac{\left(\frac{x^2 - 1}{x^2} \right)}{\left(\frac{x^2 + 1}{x^2} \right)} \rightarrow \frac{1}{1} = 1 \quad \Rightarrow \quad \text{So } \lim_{x \rightarrow \infty} \frac{x^2 - 1}{x^2 + 1} = \frac{1}{1} = 1$$

Definition: Let f be a function defined on some interval (a, ∞)

Then we write

$$\lim_{x \rightarrow \infty} f(x) = L$$

if the values of $f(x)$ can be made arbitrarily close to L by taking x sufficiently large.

We can also evaluate limits as x goes to negative infinity

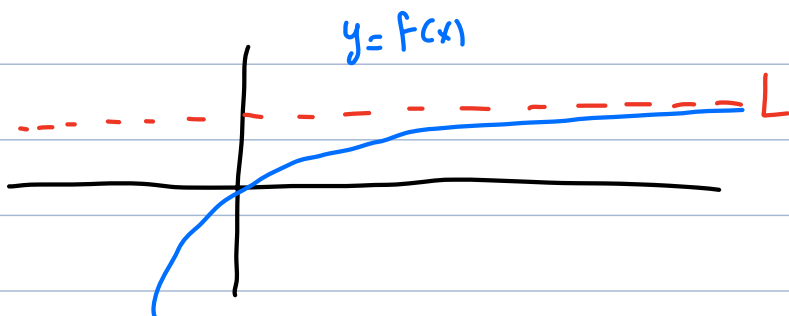
Definition: Let f be a function defined on some interval $(-\infty, a)$

Then we write

$$\lim_{x \rightarrow -\infty} f(x) = L$$

if the values of $f(x)$ can be made arbitrarily close to L by taking x sufficiently negatively large.

Picture.



$$\lim_{x \rightarrow \infty} f(x) = L$$

Definition: The line $y = L$ is called a horizontal asymptote of the curve $y = f(x)$ if

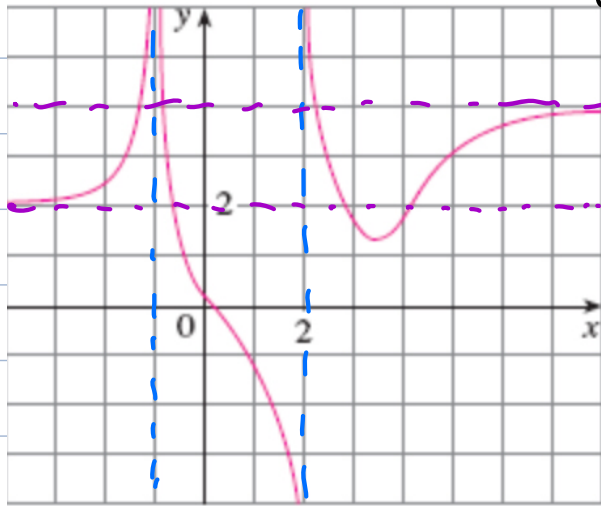
$$\lim_{x \rightarrow \infty} f(x) = L$$

$$\text{OR} \quad \lim_{x \rightarrow -\infty} f(x) = L$$

Terminology we distinguish between "infinite limits" ($\lim_{x \rightarrow a} f(x) = \infty$) and "limits at infinity" ($\lim_{x \rightarrow \infty} f(x) = L$)

vertical and
horizontal
↓

Example: Find the infinite limits, limits at infinity, and asymptotes for the function f whose graph is shown below:



$x = -1$

$x = 2$

VA: $x = -1, x = 2$

HA: $y = 2, y = 4$

infinite limits

- $\lim_{x \rightarrow -1} f(x) = \infty$

- $\lim_{x \rightarrow 2} f(x) = -\infty$

- $\lim_{x \rightarrow 2^+} f(x) = \infty$

limits at infinity:

- $\lim_{x \rightarrow \infty} f(x) = 4$

- $\lim_{x \rightarrow -\infty} f(x) = 2$

Important The limit laws we learned earlier still apply if we replace $\lim_{x \rightarrow a} f(x)$ with $\lim_{x \rightarrow \infty} f(x)$

Theorem: • If $r > 0$, then

$$\lim_{x \rightarrow \infty} \frac{1}{x^r} = 0$$

(idea: x^r grows arbitrarily large $\Rightarrow \frac{1}{x^r}$ goes to 0)

• If $r > 0$ and x^r is defined for negative numbers* then

$$\lim_{x \rightarrow -\infty} \frac{1}{x^r} = 0$$

* $r = \frac{n}{m}$ where m is odd

x^r not always defined for negative x
e.g. $(-1)^{1/2}$ undefined

Example: Evaluate

$$\lim_{x \rightarrow \infty} \frac{3x^2 - x - 2}{5x^2 + 4x + 1}$$

Intuition: Focus on dominant terms

$$\begin{aligned} 3x^2 - x - 2 &\approx 3x^2 \\ 5x^2 + 4x + 1 &\approx 5x^2 \end{aligned} \Rightarrow \frac{3x^2 - x - 2}{5x^2 + 4x + 1} \approx \frac{3x^2}{5x^2} = \frac{3}{5}$$

Actual solution: idea: divide top and bottom by dominant term (in this case x^2)

$$\lim_{x \rightarrow \infty} \frac{3x^2 - x - 2}{5x^2 + 4x + 1} = \lim_{x \rightarrow \infty} \frac{\frac{3x^2 - x - 2}{x^2}}{\frac{5x^2 + 4x + 1}{x^2}} = \lim_{x \rightarrow \infty} \frac{\frac{3x^2}{x^2} - \frac{x}{x^2} - \frac{2}{x^2}}{\frac{5x^2}{x^2} + \frac{4x}{x^2} + \frac{1}{x^2}}$$

$$= \lim_{x \rightarrow \infty} \left(\frac{3 - \frac{1}{x} - \frac{2}{x^2}}{5 + \frac{4}{x} + \frac{1}{x^2}} \right) = \frac{3 - \lim_{x \rightarrow \infty} \frac{1}{x} - 2 \lim_{x \rightarrow \infty} \frac{1}{x^2}}{5 + 4 \lim_{x \rightarrow \infty} \frac{1}{x} + \lim_{x \rightarrow \infty} \frac{1}{x^2}}$$

by limit laws

$$= \frac{3 - 0 - 2 \cdot 0}{5 + 4 \cdot 0 + 0}$$

$$= \boxed{\frac{3}{5}}$$

(you would get same result as $x \rightarrow -\infty$)

Example Find the horizontal and vertical asymptotes of the graph of

$$f(x) = \frac{\sqrt{2x^2+1}}{3x-5}$$

Intuition: focus on dominant terms

$$f(x) \approx \frac{\sqrt{2x^2}}{3x} = \frac{\sqrt{2} \cdot \sqrt{x^2}}{3x} = \frac{\sqrt{2} \cdot |x|}{3x}$$

$$\text{if } x \rightarrow \infty \text{ then } x > 0, \text{ so } \frac{\sqrt{2} x}{3x} = \frac{\sqrt{2}}{3}$$

$$\text{if } x \rightarrow -\infty \text{ then } x < 0 \text{ so } \frac{\sqrt{2} \cdot (-x)}{3x} = -\frac{\sqrt{2}}{3}$$

Formal solution:

idea: divide top and bottom by dominant term (in this case) x

note: $\sqrt{x^2} = |x|$, so compare to x

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{\sqrt{2x^2+1}}{3x-5} &= \lim_{x \rightarrow \infty} \frac{\frac{\sqrt{2x^2+1}}{x}}{\frac{3x-5}{x}} \quad \left\{ \begin{array}{l} \frac{\sqrt{2x^2+1}}{\sqrt{x^2}} = \sqrt{\frac{2x^2+1}{x^2}} \quad \left(\begin{array}{l} \text{if } x > 0 \\ x = \sqrt{x^2} \end{array} \right) \end{array} \right. \\ &= \lim_{x \rightarrow \infty} \frac{\sqrt{\frac{2x^2}{x^2} + \frac{1}{x^2}}}{\frac{3x}{x} - \frac{5}{x}} = \lim_{x \rightarrow \infty} \frac{\sqrt{2 + \frac{1}{x^2}}}{3 - \frac{5}{x}} \end{aligned}$$

by limit laws $\rightarrow$

$$\frac{\sqrt{2 + \lim_{x \rightarrow \infty} \frac{1}{x^2}}}{3 - \lim_{x \rightarrow \infty} \frac{5}{x}} = \boxed{\frac{\sqrt{2}}{3}}$$

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{2x^2+1}}{3x-5} = \lim_{x \rightarrow -\infty} \frac{\frac{\sqrt{2x^2+1}}{x} \cdot -\sqrt{x^2}}{\frac{3x-5}{x}}$$

$$\begin{cases} \text{when } x < 0 \\ x = -\sqrt{x^2} \end{cases}$$

$$= \lim_{x \rightarrow -\infty} \frac{-\sqrt{\frac{2x^2+1}{x^2}}}{\frac{3x-5}{x}} =$$

$$\lim_{x \rightarrow -\infty} \frac{-\sqrt{\frac{2x^2}{x^2} + \frac{1}{x^2}}}{\frac{3x}{x} - \frac{5}{x}} = \lim_{x \rightarrow -\infty} \frac{-\sqrt{2 + \frac{1}{x^2}}}{3 - \frac{5}{x}}$$

$$\frac{-\sqrt{2 + \lim_{x \rightarrow -\infty} \frac{1}{x^2}}}{3 - \lim_{x \rightarrow -\infty} \frac{5}{x}} = \boxed{\frac{-\sqrt{2}}{3}}$$

$$\text{HA: } y = \frac{\sqrt{2}}{3}, \quad y = -\frac{\sqrt{2}}{3}$$

$$\text{VA: } f(x) = \frac{\sqrt{2x^2+1}}{3x-5}$$

division by 0 when $3x-5=0 \Rightarrow x = \frac{5}{3}$

$$\lim_{x \rightarrow \frac{5}{3}^-} \frac{\sqrt{2x^2+1}}{3x-5} = \boxed{-\infty}$$

$$\frac{\sqrt{2x^2+1}}{3x-5} \left\{ \begin{array}{l} + \\ - \end{array} \right. = -$$

$$x < \frac{5}{3} \Rightarrow 3x-5 < 0$$

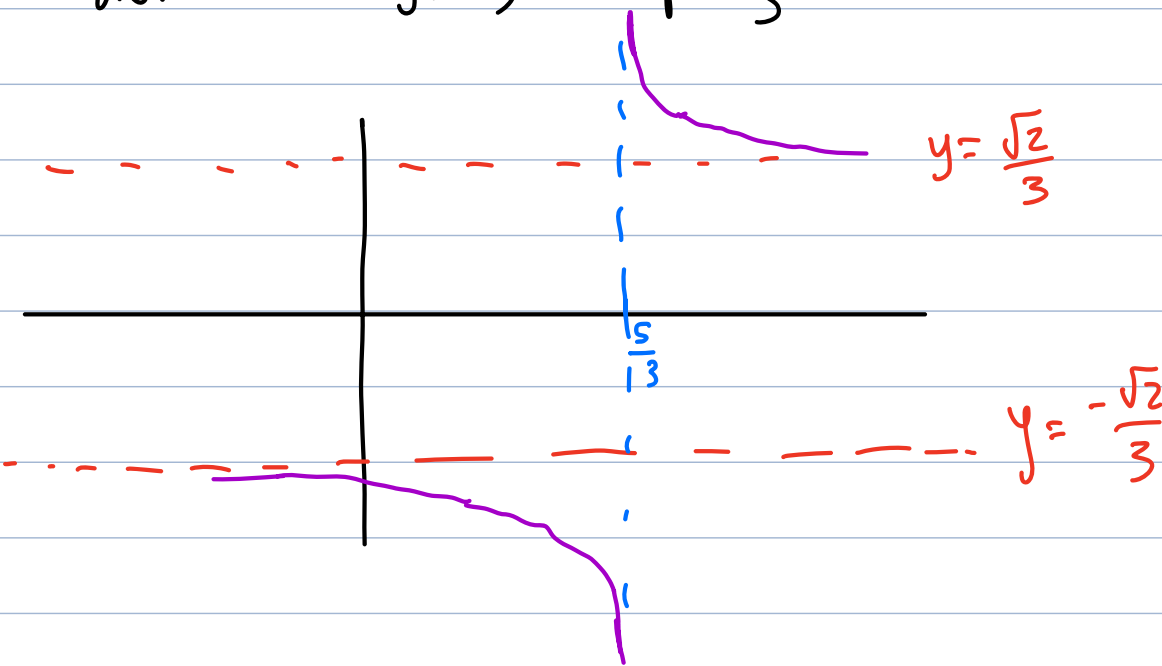
$$\lim_{x \rightarrow \frac{5}{3}^+} \frac{\sqrt{2x^2+1}}{3x-5} = \boxed{\infty}$$

$$x > \frac{5}{3} \Rightarrow 3x-5 > 0$$

$$\frac{\sqrt{2x^2+1}}{3x-5} \} + = +$$

sketch:

domain everything except $\frac{5}{3}$



Example Compute $\lim_{x \rightarrow \infty} (\sqrt{x^2+2x} - x)$

note: this limit is of the form " $\infty - \infty$ " which is indeterminate

Caution: intuition of "dominant term" unreliable with subtraction.

Wrong answer:

$$\sqrt{x^2+2x} \approx \sqrt{x^2} = x$$

$$\sqrt{x^2+2x} - x \approx x - x = 0$$

idea: multiply and divide by conjugate

$$\lim_{x \rightarrow \infty} (\sqrt{x^2+2x} - x) = \lim_{x \rightarrow \infty} \frac{(\sqrt{x^2+2x} - x) \cdot (\sqrt{x^2+2x} + x)}{\sqrt{x^2+2x} + x}$$

diff
of squares

$$= \lim_{x \rightarrow \infty} \frac{(\sqrt{x^2+2x})^2 - x^2}{\sqrt{x^2+2x} + x} = \lim_{x \rightarrow \infty} \frac{\cancel{x^2} + 2x - \cancel{x^2}}{\sqrt{x^2+2x} + x}$$

$$= \lim_{x \rightarrow \infty} \frac{2x}{\sqrt{x^2+2x} + x}$$

now multiply and divide
by dominant term (x)

$$= \lim_{x \rightarrow \infty} \frac{\frac{2x}{x}}{\frac{\sqrt{x^2+2x}}{x} + \frac{x}{x}} = \lim_{x \rightarrow \infty} \frac{2}{\frac{\sqrt{x^2+2x}}{\sqrt{x^2}} + 1} \quad \left(x = \sqrt{x^2} \text{ when } x > 0 \right)$$

$$= \lim_{x \rightarrow \infty} \frac{2}{\sqrt{\frac{x^2+2x}{x^2}} + 1} = \lim_{x \rightarrow \infty} \frac{2}{\sqrt{1 + \frac{2}{x}} + 1} = \frac{2}{\sqrt{1 + \lim_{x \rightarrow \infty} \frac{2}{x}} + 1}$$

$$= \frac{2}{\sqrt{1+1}} = \frac{2}{2} = \boxed{1}$$

Some other limits at infinity

Evaluate the following limits (or DNE if they do not exist):

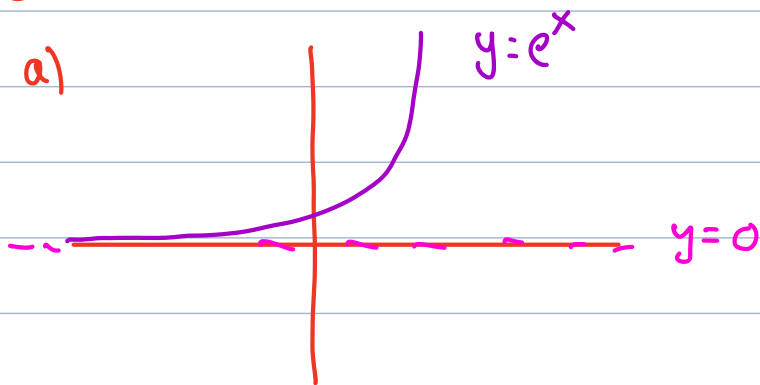
a) $\lim_{x \rightarrow -\infty} e^x$

b) $\lim_{x \rightarrow 0^-} e^{1/x}$

c) $\lim_{x \rightarrow \infty} \tan^{-1}(x)$

d) $\lim_{x \rightarrow \infty} \sin(x)$

Solutions:



$$\lim_{x \rightarrow -\infty} e^x = 0$$

b) as $x \rightarrow 0^-$, $\frac{1}{x} \rightarrow -\infty$, $e^{\frac{1}{x}} \rightarrow 0$

$$\lim_{x \rightarrow 0^-} e^{1/x} = 0$$

Alternatively, use substitution

$$\lim_{x \rightarrow 0^-} e^{1/x}$$

let $u = \frac{1}{x}$.

as $x \rightarrow 0^-$, $u = \frac{1}{x} \rightarrow -\infty$

$$\left. \begin{array}{l} \text{let } u = \frac{1}{x} \\ \text{as } x \rightarrow 0^-, u = \frac{1}{x} \rightarrow -\infty \end{array} \right\} \lim_{x \rightarrow 0^-} e^{1/x} = \lim_{u \rightarrow -\infty} e^u = \boxed{0}$$

Example find limits at infinity of $f(x) = \frac{e^x + 1}{e^x + 2}$

• $\lim_{x \rightarrow \infty} \frac{e^x + 1}{e^x + 2}$

Substitution: let $u = e^x$
As $x \rightarrow \infty$, $u = e^x \rightarrow \infty$

$$\lim_{u \rightarrow \infty} \frac{u+1}{u+2}$$

divide top and bottom by dominant term (u)

$$= \lim_{u \rightarrow \infty} \frac{\frac{u+1}{u}}{\frac{u+2}{u}} = \lim_{u \rightarrow \infty} \frac{\frac{u}{u} + \frac{1}{u}}{\frac{u}{u} + \frac{2}{u}} = \lim_{u \rightarrow \infty} \frac{1 + \frac{1}{u}}{1 + \frac{2}{u}}$$

$$= \frac{1 + \lim_{u \rightarrow \infty} \frac{1}{u}}{1 + \lim_{u \rightarrow \infty} \frac{2}{u}} = \frac{1}{1} = \boxed{1}$$

• $\lim_{x \rightarrow -\infty} \frac{e^x + 1}{e^x + 2}$

Substitution: let $u = e^x$
As $x \rightarrow -\infty$, $u = e^x \rightarrow 0$

$$\lim_{u \rightarrow 0} \frac{u+1}{u+2} = \text{direct subst. } u=0$$

$$\boxed{\frac{1}{2}}$$

Infinite limits at infinity

We can combine the concepts of "infinite limits" and "limits at infinity" to get "infinite limits at infinity"

Example Find the infinite limits

$$\lim_{x \rightarrow \infty} x^3 \text{ and } \lim_{x \rightarrow -\infty} x^3$$

(answer should be of the form ∞ or $-\infty$)

$$\lim_{x \rightarrow \infty} x^3 = \infty$$

$$\lim_{x \rightarrow -\infty} x^3 = -\infty$$

(negative cubed is negative)

Example Find $\lim_{x \rightarrow \infty} (x^2 - x)$ } this is of form " $\infty - \infty$ " indeterminate

intuition: x^2 grows a lot faster than x

$$\text{so } x^2 \gg x \Rightarrow x^2 - x \rightarrow \infty$$

Can think about the graph



More formally:

$$\lim_{x \rightarrow \infty} (x^2 - x) = \lim_{x \rightarrow \infty} \overset{\infty}{\uparrow} x \overset{\infty}{\uparrow} (x-1) = \infty$$

(" $\infty \cdot \infty = \infty$ ")

Example Find $\lim_{x \rightarrow \infty} \frac{x^2+x}{3-x}$ } $\frac{\infty}{-\infty}$ indeterminate

Intuition: focus on dominant terms

$$\frac{x^2+x \approx x^2}{3-x \approx -x} = -x \rightarrow -\infty$$

More precise explanation:

divide top and bottom by dominant term of denominator
(in this case, x)

$$\lim_{x \rightarrow \infty} \frac{x^2+x}{3-x} = \lim_{x \rightarrow \infty} \frac{\frac{x^2+x}{x}}{\frac{3-x}{x}} = \lim_{x \rightarrow \infty} \frac{\boxed{x+1} \rightarrow \infty}{\boxed{\frac{3}{x}-1} \rightarrow 0-1 = -1}$$
$$= -\infty$$

$$\text{"} \frac{\infty}{-c} = -\infty \text{"}$$